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The Illegal Wildlife Trade in the Brazilian Amazon: A Systematic Review of Patterns, Routes, and Drivers

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Abstract

The Amazon has a long history of wildlife exploitation, which continues to this day through the illegal wildlife trade (IWT). Although research conducted in the region provides relevant information at the local level, gaps remain in broader regional views of IWT patterns and trends. We conducted a systematic literature review to characterize IWT in the Brazilian Amazon, identifying trade routes and assessing the factors that drive this activity. Following the PRISMA-Eco-Evo protocol, we evaluated 34 scientific articles. We recorded the illegal trade of 137 species, representing all vertebrate groups, of which 25.5% are included in threatened categories of the ICMBio/MMA (2018). We identified seven purposes for wildlife use, with human consumption being the most prominent. Most articles (88%) documented trade occurring in publicly accessible locations, such as fairs and markets. Trade routes encompassed both national and international scales. We identified cultural, biological, socioeconomic, and environmental factors as drivers of IWT. Despite being illegal, IWT remains widespread in the Amazon, likely due to high demand and weak enforcement. The identified geographic and thematic gaps highlight the need to expand the scope of research, particularly in remote and border areas. Furthermore, the observed drivers are similar to those reported in other regions of Brazil and worldwide, indicating a convergence of socioenvironmental motivations and vulnerabilities associated with the trade. Our study highlights the need to strengthen education and awareness strategies to reduce the demand for wildlife, in addition to enhancing enforcement and efforts to combat IWT in the Amazon.

Keywords Amazon conservation · Animal trade · Brazil · Game meat · Wildlife trafficking

Introduction

The illegal wildlife trade (IWT), that is, the unauthorized sale of animals and their products for a wide range of uses (Sas-Rolfes et al., 2019), is a growing global conservation concern (Phelps et al., 2016). Although IWT is often used in the literature to refer to the entire trafficking chain (Costa et al., 2017; Sas-Rolfes et al., 2019), this study focuses specifically on the final stage, that is, the commercialization stage, defined here as the trade of wildlife and their products. This form of illegal and unsustainable trade not only threatens the local survival of many animal species but also contributes to the spread of invasive species (García-Díaz et al., 2017), the occurrence of zoonotic diseases (Brookes et al., 2022), and the disruption of ecological processes such as seed dispersal and pollination (Dirzo et al., 2014; Hughes et al., 2022). In addition to its implications for biodiversity, IWT also impacts society, particularly people who depend on wildlife for food security and livelihoods in tropical rural areas (Nielsen et al., 2018).

Understanding the various aspects related to IWT is essential for proposing measures to counter this activity. Socioeconomic factors such as poverty or unfavorable social conditions (Scheffers et al., 2019), as well as

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low education levels and high unemployment rates (Alves et al., 2013; Harrison et al., 2013), are frequently cited as important drivers associated with wildlife trade. For animals traded as pets, biological traits related to appearance and behavior are recognized as drivers, such as coloration in amphibians (Werling et al., 2025) and singing ability in birds (Silva et al., 2022). Additionally, cultural factors drive the use and illegal trade of wildlife (Awoyemi et al., 2022). In Southeast Asia, for example, the trade in non-human primates (genera *Loris* and *Nycticebus*) is strongly influenced by folklore and traditional beliefs in the healing and mystical powers of these species (Nekaris et al., 2010).

Brazil is a megadiverse country where IWT occurs in all regions (Destro et al., 2012), despite the existence of laws that prohibit and penalize both the capture and trade of wild fauna specimens (Brazil, 1967, 1998). At the national level, most animals are traded as pets, primarily birds and, to a lesser extent, reptiles and mammals (Destro et al., 2012; Azeredo & Alves, 2024). Moreover, there is also demand for wildlife for human consumption, particularly in commercial contexts, in the form of wild meat (van Vliet et al., 2015a); for medicinal purposes, in the production of natural remedies (Alves et al., 2010; Melo et al., 2014); and for magical-religious purposes, during spiritual practices and folk rituals (Alves et al., 2012). Taken together, all these forms of use contribute to the demand for wildlife in Brazil and the consequent illegal exploitation of animals.

The Amazon biome, which accounts for approximately 50% of the Brazilian territory (IBGE, 2024) and for much of the country's biodiversity, has a long history of wildlife exploitation (Antunes et al., 2014) that extends to the present day through the illegal trade of animals. In the Amazon region, there is high demand for wildlife for human consumption (Parry et al., 2014; el Birzi et al., 2020), particularly medium- and large-bodied mammals (e.g., *Cuniculus paca*, *Tapirus terrestris*; Mendes, 2020), reptiles, and fishes, which are often traded in fairs and public markets (van Vliet et al., 2015a; Charity & Ferreira, 2020). The pet trade is also recurrent, with birds such as the chestnut-bellied seed finch (*Sporophila angolensis*; Farias et al., 2019), fishes such as the cardinal tetra (*Paracheirodon axelrodi*) and freshwater stingrays (*Potamotrygon spp.*; Beltrão et al., 2021). The state of Amazonas is also part of the main international trade route for ornamental fishes in Brazil (Charity & Ferreira, 2020; Beltrão et al., 2021; Borges et al., 2021). In addition, the trade of products and animals for medicinal (Pinto & Maduro, 2003; Alves et al., 2010), religious, and mystical purposes (Alves et al., 2012), such as for carrying out folk rituals and spiritual practices, also represents a source of wildlife demand in the region.

Since it is an illegal activity, investigating and monitoring IWT represents a considerable challenge (Phelps et al., 2016), which becomes even more complex in a region as vast and biodiverse as the Amazon. The body of published literature on IWT in the region provides important information on the species traded, the types of products most in demand, points of sale, drivers of the illegal trade, trade flows, and actors involved in the trade chain, among other aspects relevant to understanding how this activity operates (e.g., Baía Júnior et al., 2010; van Vliet et al., 2015). However, these studies present data that provide isolated and/or locally focused information, lacking a comprehensive and integrated view of the illegal trade in the biome. Such a perspective is of utmost importance because collecting data at a regional scale provides information on: (i) knowledge gaps, that is, areas with insufficient information; (ii) the most frequently traded animals and the most threatened species; and (iii) general factors that foster trade. Furthermore, a study that takes a broader perspective to reveal the patterns and trends of this activity provides a strong tool for developing a cohesive conservation plan for the region.

Considering this, this study provides the first systematic literature review on IWT in the Brazilian Amazon. Our objectives were to characterize the trade and the wildlife involved, identify trade routes, and assess the factors driving IWT in Brazil's largest biome. We hope that this work will contribute to a broader understanding of this trade, providing a scientific basis to inform discussions on the topic, thereby supporting the development of approaches to combat IWT and conservation strategies aimed at preserving Amazonian biodiversity. We also hope that our data will help elucidate regional trade patterns and trends, as well as identify potential research gaps that merit further investigation.

Materials and Methods

This systematic review followed the recommendations of the PRISMA-EcoEvo protocol (O'Dea et al., 2021). Between October and December 2023, we conducted preliminary searches to identify the most appropriate descriptors to be used as keywords. Between January and April 2024, we carried out the searches in the Web of Science, Scopus, and SciELO databases using a set of 18 keyword combinations related to IWT in the Amazon (see Supplementary Material S1). In May 2026, we conducted a complementary search using the same databases, keyword combinations, and criteria, covering the period from April 2024 to May 2026; no new eligible studies were identified. Title and abstract screening were performed by MCL and MFDF, with discrepancies resolved through discussion. Full-text reading and data extraction were performed by MCL.

We selected only articles written in English and Portuguese, published in scientific journals, and peer-reviewed. We chose to exclude the gray literature to ensure methodological consistency and the reliability of the data included in the review, as the absence of peer review limits the ability to assess the accuracy and quality of reported findings. This decision was based on preliminary exploratory searches, which indicated that much of the gray literature on this topic lacked adequate taxonomic identification, quantitative data, and precise study location, and was often outside the scope of this review. Our inclusion criteria were articles: (1) related to IWT (whether studies on public markets, street fairs, research with animal traders or commercial hunters); (2) conducted between 1967 (the year of enactment of the Wildlife Protection Law – Law No. 5,197/1967, Brazil, 1967) and May 2026 (the date of our most recent database search); and (3) fully or partially carried out within the Brazilian Amazon territory. For studies conducted partially outside the Amazon but including data from this region, we used only the information related to locations within the biome's boundaries.

Our exclusion criteria were: (1) articles that did not provide taxonomic information on the traded animals (at least to the genus level); (2) articles explicitly related to the use of wildlife for subsistence, such as those conducted in Indigenous communities (Brazil, 1973) and (3) articles addressing other wildlife-related crimes (e.g., hunting, transport, possession), such as those analyzing seizure data of wild animals. Studies lacking taxonomic identification were excluded, as the absence of such information limits the ability to assess trade dynamics and prevents linking records to conservation status and ecological relevance. The exclusion of studies conducted in Indigenous communities reflects the definition of IWT adopted in this review, which focuses on forms of illegal wildlife trade involving commercial exchange. As the subsistence use of wildlife by these populations is legally recognized in Brazil (Brazil, 1973), such practices fall outside the scope of this study. We acknowledge, however, that this represents a specific operational definition of IWT, and that Indigenous contexts may intersect with commercial trade networks in ways not captured by this review. Similarly, articles addressing other wildlife-related crimes (e.g., illegal possession or transport), including those based on seizure data, were excluded because they encompass a broader range of offenses that do not necessarily reflect trade dynamics and therefore fall outside the scope of IWT as defined in this review.

Initially, we evaluated the articles by reading their titles and abstracts and selected those that met our inclusion criteria. Subsequently, we performed a full-text reading of the articles and excluded those that did not meet our standards (see Supplementary Material S2 for more details). Then, using the Snowballing method (Wohlin, 2014), we checked the reference lists of all selected articles to identify potential studies for inclusion in the review. We read these articles in full and included those that met the criteria of our study.

The searches, applying the keywords in the databases, returned a total of 3,354 articles (Fig. 1). After reading the titles and abstracts and applying the inclusion criteria, we selected 207 articles. As the searches were conducted manually across multiple databases, duplicates ($n = 142$) were identified and removed after this initial screening, resulting in 65 unique articles. Among these, we eliminated 47 after applying the exclusion criteria, resulting in a sample of 18 articles selected for the review. We then checked the reference lists of these studies (Snowballing) and included 16 additional articles that met our criteria, for a final total of

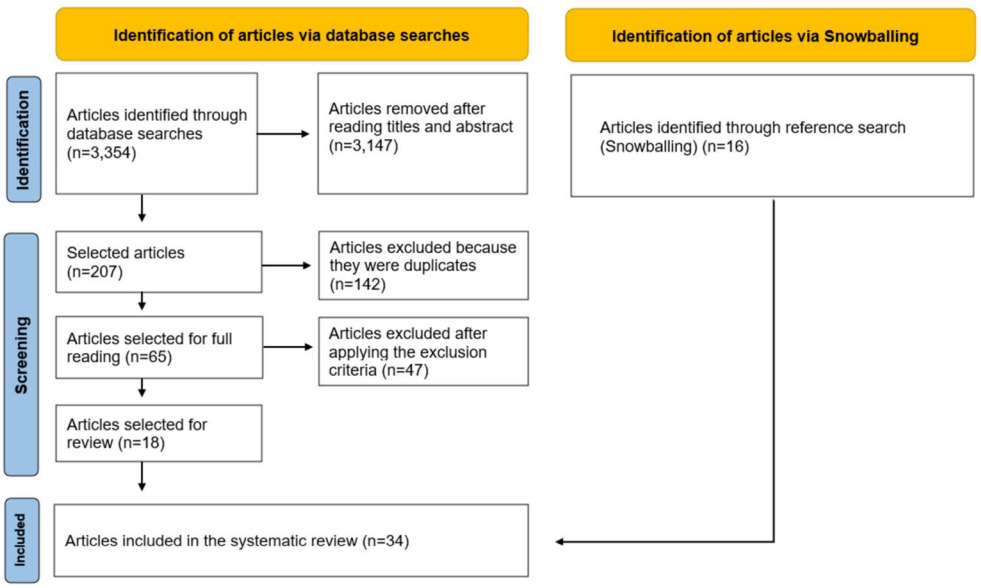


Fig. 1 Flowchart of article selection, based on the PRISMA-EcoEvo protocol, illustrating the systematic identification, screening, and inclusion of articles for the systematic review on IWT in the Brazilian Amazon

34 articles included in the review. (see the complete list of articles in Supplementary Material S3).

To examine the geographical distribution of the study sites, characterize the trade and the traded fauna, identify trade routes, and assess the factors driving IWT in the Brazilian Amazon, we manually extracted the corresponding data from each article, whether from the text, figures, tables, or Supplementary materials provided by the authors. We collected the following information: title, authors, study location (data collection site), traded species, products or animal parts traded, place or mode of trade, and intended use. Furthermore, when available in the articles, we extracted information regarding the trade routes and drivers of IWT. We classified the drivers into four categories: cultural (beliefs, traditions, and cultural practices), biological (characteristics inherent to the animals), socioeconomic (price of animal products), and environmental (seasonality) (see inclusion criteria for drivers in the Supplementary Material S5). These categories were defined through an interpretative synthesis conducted by the authors, based on similarities among drivers reported in the primary studies and informed by existing literature (Hughes et al., 2022; Destro et al., 2019). Individual studies could be assigned to more than one category when multiple drivers were identified. Data on species or animal products that are legally allowed for use, such as domestic animals (e.g., cattle, pigs) and legally traded fish, were not included in the review. To determine the species of fish for which trade is prohibited, we followed the provisions of MMA Ordinance No. 445/2014 (Brazil, 2014) and the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES, 1973). We checked and updated the scientific nomenclature of the species according to the Catalogue of Life (COL) database (https://www.catalogueoflife.org/), the Reptile Database (http://www.reptile-database.org/) and the Brazilian Ornithological Records Committee (https://www.cbro.org.br/listas/) (see the taxonomic standardization in Supplementary Material S6). We assessed the conservation status of each species according to the Red Book of Threatened Brazilian Fauna (ICMBio/MMA, 2018).

The data were tabulated in Microsoft Excel, and we compiled lists of recorded species, intended uses, and traded parts and products. We produced maps of illegal trade routes and the geographical distribution of the articles using QGIS software version 3.36.1. We performed descriptive analyses of frequency and percentage to evaluate the geographical distribution of the articles, characterize the trade and the traded animals, and assess the trade routes. We calculated percentages based on the number of articles included in the review ($n=34$) and on the total number of species recorded in the articles ($n=137$). To assess the drivers associated with wildlife trade, we calculated the percentage based on the number of articles that mentioned these factors ($n=18$). For calculations of intended uses and places of trade of fauna, the percentages exceeded 100% because some articles reported the use of the same species for two or more purposes, or trade occurring in two or more locations. The same applies to analyses of the geographical distribution of the articles and the categories of IWT drivers.

Results

Geographical Distribution of Study Sites on IWT in the Brazilian Amazon

Among the nine states of the Amazon biome, five had studies addressing the illegal trade of wildlife: Amazonas, Amapá, Pará, Maranhão, and Roraima (Fig. 2). More than half of the studies were conducted entirely or partially in the state of Pará ($n=23$, 67.6%), most of them in the capital Belém ($n=14$, 41.1%). Amazonas ranked second ($n=13$, 38.2%), with emphasis on the Alto Solimões mesoregion ($n=7$, 20.5%), followed by Maranhão ($n=5$, 14.7%), Amapá ($n=1$, 2.9%), and Roraima ($n=1$, 2.9%). Most of the studies ($n=30$, 88.2%) were conducted in municipalities located along the banks of the major rivers of the Amazon region. Four studies (11.8%) were conducted in border areas with neighboring countries.

Locations and Forms of Illegal Wildlife Trade in the Brazilian Amazon

Regarding places of trade, most of the articles ($n=30$, 88.2%) reported that trade occurred in physical locations with public access, such as street markets ($n=26$, 76.4%), religious goods stores ($n=4$, 11.7%), among others (Fig. 3). Beyond these fixed locations, we also recorded trade occurring through middlemen ($n=10$, 29.4%), through orders ($n=5$, 14.7%), and through street vendors ($n=2$, 5.8%). Middlemen are individuals who act as intermediaries between hunters and the final consumers or animal traders (Sas-Rolfes et al., 2019). In trade through orders, consumers request animals directly from hunters or traders and later receive their order (van Vliet et al., 2015). Street vendors circulate while carrying animals to offer to buyers (Farias et al., 2019).

Characterization of Wildlife Illegally Traded in the Brazilian Amazon (Species, Purposes of Use, Conservation Status, and Traded Products or Parts)

We identified a total of 137 wild animal species being illegally traded in the Amazon region (Table 1), with birds

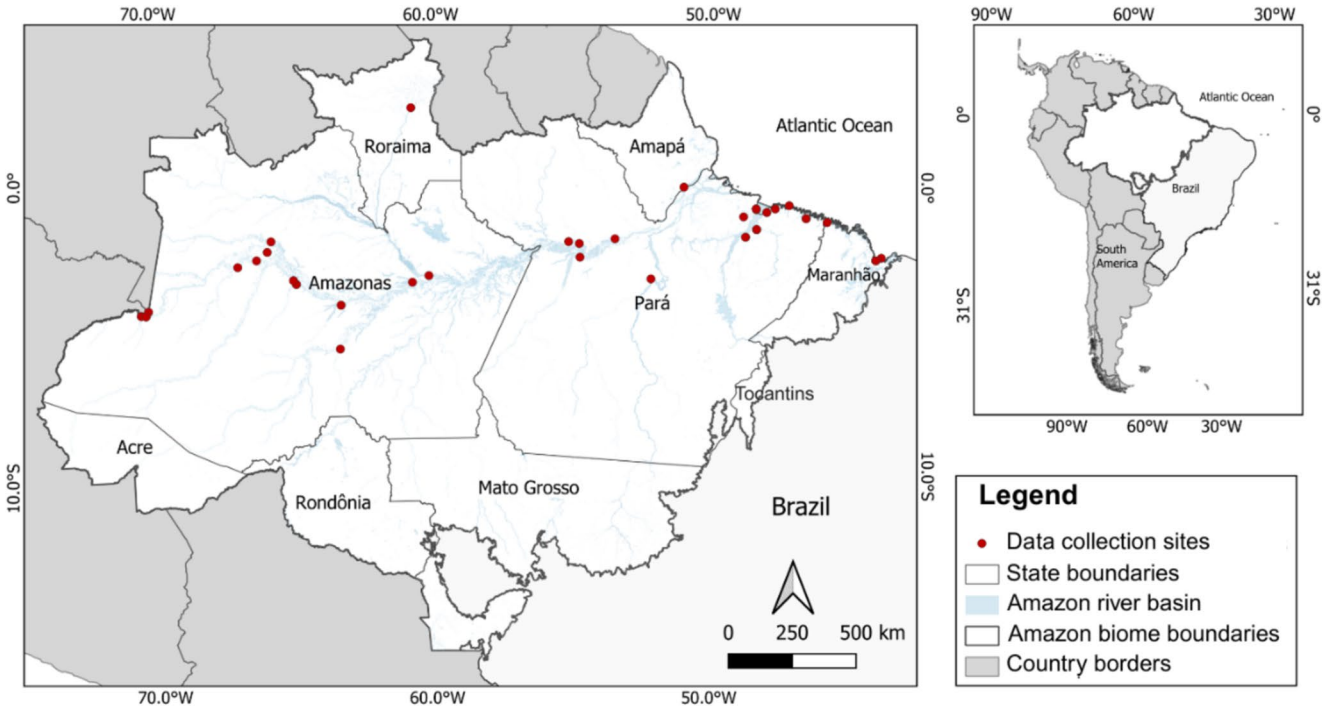


Fig. 2 Geographic distribution of data collection sites (red dots) across Brazilian states within the Amazon biome, based on articles included in the systematic review ($n=34$)

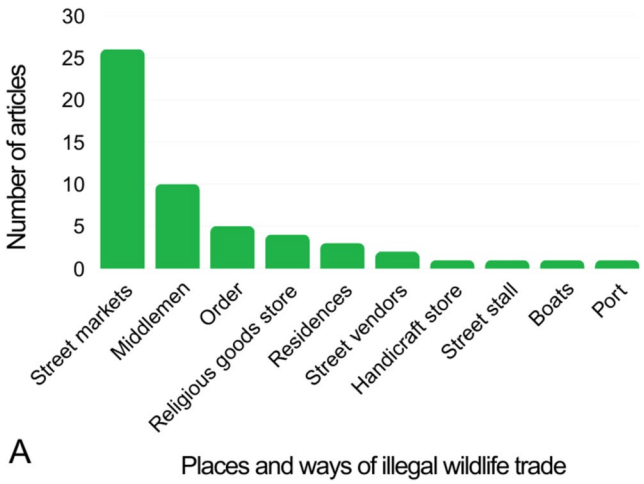


Fig. 3 Locations and forms of IWT in the Brazilian Amazon, as reported in the reviewed articles ($n=34$)

being the most represented group ($n=50$, 36.4%), followed by mammals ($n=39$, 28.4%), reptiles ($n=26$, 18.9%), fishes ($n=21$, 15.3%), and amphibians ($n=1$, 0.7%) (Fig. 4A). Additionally, 14 records of animals were classified only at the genus level (*Aburria*, *Bothrops*, *Bufo*, *Crax*, *Crypturellus*, *Dasyprocta*, *Dasypus*, *Leptodactylus*, *Mazama*, *Micrurus*, *Mitu*, *Penelope*, *Sporophila*, *Tolypeutes*).

We identified seven purposes of use for the 137 recorded species, with food ($n=51$, 37.2%), magical-religious ($n=50$, 36.4%), and medicinal ($n=44$, 32.1%) being the most prominent (Fig. 4B). Pet trade accounted for 25 species (18.2%), while ornamental ($n=3$, 2.1%), laboratory ($n=3$, 2.1%; specifically primates used in laboratory testing), and cosmetic uses ($n=1$, 0.7%) were less frequent. Some species were reported for two or more purposes, such as the black caiman (*Melanosuchus niger*), which is used for food, medicinal, and magical-religious practices.

Regarding the conservation status of the recorded species, 35 (25.5%) are classified as threatened according to the ICMBio/MMA (2018) categories (see Table 1). Critically Endangered (CR) species were predominantly fishes ($n=11$), with one bird, while Endangered (EN) species comprised two fishes and two mammals (Fig. 5). Among the Vulnerable (VU) species, mammals were the most represented ($n=12$), followed by fishes ($n=6$) and one bird.

Regarding the animal parts or products traded, we found 26 items traded, for example: meat, fat and eggs (see Supplementary Material S4 for the complete list). Certain animals were traded whole and slaughtered (e.g., *Cuniculus paca*; Baía Júnior et al., 2010) or in portions, whether as fresh meat (e.g., tapir, *Tapirus terrestris*; van Vliet et al., 2014), chilled meat (e.g., deer, *Mazama americana*; Baía Júnior et al., 2010), or salt-preserved meat (e.g., caiman, *Caiman crocodilus*; da Silveira & Thorbjarnarson, 1999). Others were traded whole and alive, including turtles for

Table 1 List of wildlife species traded in the Brazilian Amazon, according to the reviewed literature. Conservation status follows the ICMBio/MMA classification (2018): CR: critically endangered; EN: endangered; VU: vulnerable; LC: least concern; DD: data deficient; NT: near threatened; NL: not listed

GROUP Family	Species or genus	Common name in Portuguese	Conservation status	Purpose of use
FISHES				
Carcharhinidae	*Carcharhinus porosus*	Cação-mole	CR	Food
	Carcharhinus acronotus	Tubarão-de-nariz-preto	VU	Food
	Carcharhinus falciformis	Tubarão-seda	CR	Food
	Carcharhinus leucas	Tubarão-cabeça-chata	VU	Food
	Carcharhinus limbatus	Tubarão-galha-preta	VU	Food
	Carcharhinus plumbeus	Tubarão-galhudo	CR	Food
	Isogomphodon oxyrhynchus	Cação-quati	CR	Food
Ginglymostomatidae	*Ginglymostoma cirratum*	Tubarão-lixa, Cação	VU	Food
Dasyatidae	*Hypanus americanus*	Arraia-lixa	VU	Food
Loricariidae	*Hypancistrus zebra*	Cascudo-zebra	CR	Pet
Myliobatidae	*Rhinoptera brasiliensis*	Arraia-ticonha	CR	Food
Pristidae	*Pristis perotteti*	Peixe-serra	NL	Medicinal
	Pristis pectinata	Peixe-serra-dentes-pequenos	CR	Medicinal
	Pristis pristis	Peixe-serra	CR	Food
Sphyrnidae	*Sphyrna tudes*	Tubarão-martelo	CR	Food
	Sphyrna lewini	Tubarão-martelo-recortado	CR	Food
	Sphyrna mokarran	Cação-panã	EN	Food
	Sphyrna tiburo	Tubarão-martelo	CR	Food
Syngnathidae	*Hippocampus reidi*	Cavalo-marinho	VU	Medicinal, Magical-religious
Triakidae	*Mustelus higmani*	Tubarão-mustelus	LC	Food
	Mustelus canis	Cação-bico-doce	EN	Food
AMPHIBIANS				
Bufonidae	*Bufo sp.*	Sapo	NL	Medicinal
	Rhinella jimi	Sapo-cururu	LC	Medicinal
Leptodactylidae	*Leptodactylus sp.*	Sapo-gia	NL	Medicinal
REPTILES				
Alligatoridae	*Melanosuchus niger*	Jacaré-negro, Jacaré-açu	LC	Food, Medicinal, Magical-religious
	Caiman crocodilus	Jacaré-tinga	LC	Medicinal, Magical-religious
	Paleosuchus palpebrosus	Jacaré-anão	LC	Medicinal, Magical-religious
	Caiman latirostris	Jacaré-do-papo-amarelo	LC	Medicinal, Magical-religious
	Paleosuchus trigonatus	Jacaré-coroa	LC	Medicinal, Magical-religious
Boidae	*Boa constrictor*	Jibóia	LC	Medicinal, Magical-religious
	Eunectes murinus	Anaconda, Sucuri	LC	Medicinal, Magical-religious
	Corallus caninus	Cobra-verde	LC	Medicinal
Chelidae	*Chelus fimbriata*	Matamatá	NL	Food
	Phrynops geoffroanus	Cágado-de-barbicha	NL	Medicinal
Colubridae	*Leptophis ahaetulla*	Azulão-bóia	LC	Medicinal
	Spilotes pullatus	Caninana	LC	Medicinal
Dipsadidae	*Erythrolamprus sp.*	Coral-falsa	NL	Magical-religious
Elapidae	*Micrurus sp.*	Cobra-coral	NL	Medicinal, Magical-religious
Geoemydidae	*Rhinoclemmys punctularia*	Jabuti-aperema	LC	Magical-religious
Iguanidae	*Iguana iguana*	Iguana-verde	LC	Medicinal
Kinosternidae	*Kinosternon scorpioides*	Muçuã, Jurará	LC	Food
Podocnemididae	*Podocnemis expansa*	Tartaruga-da-Amazônia	NT	Food, Medicinal, Cosmetic
	Podocnemis unifilis	Tracajá	NT	Food, Magical-religious
	Podocnemis sextuberculata	Iaçá, Pitiú	NT	Food, Medicinal
	Peltocephalus dumerilianus	Tartaruga-de-cabeça-grande	LC	Food
Teiidae	*Tupinambis teguixin*	Teiú-branco	LC	Medicinal, Magical-religious
	Uranoscodon superciliosus	Tamaquaré	LC	Medicinal, Magical-religious
Testudinidae	*Chelonoidis denticulatus*	Jabuti-tinga	LC	Food, Magical-religious

Table 1 (continued)

GROUP Family	Species or genus	Common name in Portuguese	Conservation status	Purpose of use
	Chelonoidis carbonarius	Jabuti-piranga	LC	Medicinal, Magical-religious
Viperidae	*Crotalus durissus*	Cascavel	LC	Medicinal, Magical-religious
	Bothrops sp.	Jararaca	NL	Medicinal
	Lachesis muta	Surucucu	LC	Medicinal, Magical-religious
BIRDS				
Accipitridae	*Harpia harpyja*	Gavião-real	VU	Magical-religious
Anatidae	*Cairina moschata*	Pato-do-mato	LC	Food
	Dendrocygna autumnalis	Marreca-cabocla	LC	Food
Anhingidae	*Anhinga anhinga*	Biguá-tinga	LC	Food
Apodidae	*Panyptila cayennensis*	Andorinhão-estofador	LC	Magical-religious
Cardinalidae	*Cyanoloxia rothschildii*	Azulão-da-Amazônia	LC	Pet
Caprimulgidae	*Nyctidromus albicollis*	Bacurau	LC	Medicinal, Magical-religious
Cathartidae	*Coragyps atratus*	Urubu	LC	Medicinal, Magical-religious
Columbidae	*Columba livia*	Pombo-doméstico	NL	Magical-religious
Cracidae	*Crax sp.*	Mutum	NL	Food
	Mitu sp.	Mutum	NL	Food
	Penelope sp.	Jacú	NL	Food
	Aburria sp.	Jacutinga	NL	Food
	Mitu tuberosum	Mutum-cavalo	NT	Food
Cuculidae	*Crotophaga ani*	Anu-preto	LC	Medicinal, Magical-religious
	Tapera naevia	Saci, Matinta-pereira	LC	Magical-religious
Falconidae	*Falco rufigularis*	Gavião-de-coleira	LC	Magical-religious
	Herpetotheres cachinnans	Acauã	LC	Medicinal, Magical-religious
	Caracara plancus	Carcará	LC	Medicinal, Magical-religious
Fringillidae	*Euphonia cayennensis*	Gaturamo-preto	LC	Pet
	Euphonia chlorotica	Fim-fim	LC	Pet
	Euphonia violacea	Gaturamo-verdadeiro	LC	Pet
Hirundinidae	*Hirundo rustica*	Andorinha	LC	Magical-religious
Icteridae	*Cacicus cela*	Xexéu, Japiim	LC	Pet
	Icterus cayanensis	Inhapim	LC	Pet
	Icterus jamacaii	Corrupião	NL	Pet
Picidae	*Dryocopus lineatus*	Pica-pau	LC	Medicinal
Pipridae	*Pipra fasciicauda*	Uirapuru-laranja	LC	Magical-religious
	Pipra aureola	Uirapuru	LC	Medicinal
	Pipra rubrocapilla	Cabeça-encarnada	LC	Magical-religious
Ramphastidae	*Ramphastos tucanus*	Tucano	LC	Medicinal
	Pteroglossus aracari	Araçari-de-bico-branco	LC	Medicinal
	Pteroglossus inscriptus	Araçari-de-bico-riscado	LC	Medicinal
	Ramphastos vitellinus	Tucano-de-bico-preto	LC	Medicinal
Rheidae	*Rhea americana*	Ema	LC	Medicinal
Strigidae	*Glaucidium brasilianum*	Caburé	LC	Magical-religious
Thraupidae	*Saltator maximus*	Trinca-ferro, Tempera-viola	LC	Pet
	Sicalis luteola	Tipio	LC	Pet
	Sporophila americana	Coleiro-do-norte	LC	Pet
	Sporophila angolensis	Curió	LC	Pet
	Sporophila collaris	Coleiro-do-brejo	LC	Pet
	Sporophila crassirostris	Bicudinho, Bicudo-do-norte	LC	Pet
	Sporophila lineola	Bigodinho	LC	Pet
	Sporophila maximiliani	Bicudo	CR	Pet
	Sporophila minuta	Cabolinho-paraense	LC	Pet
	Sporophila plumbea	Patativa	LC	Pet
	Sporophila schistacea	Cigarrinha-do-norte	LC	Pet
	Tachyphonus rufus	Pipira-preta	LC	Pet

Table 1 (continued)

GROUP Family	Species or genus	Common name in Portuguese	Conservation status	Purpose of use
	Tangara episcopus	Sanhaço-da-Amazônia	LC	Pet
	Volatinia jacarina	Tiziu	LC	Pet
	Sporophila sp.	Papa-capim	NL	Pet
	Crypturellus sp.	Inhambu	NL	Pet
Troglodytidae	*Cyphorhinus arada*	Uirapuru-verdadeiro	LC	Magical-religious
Turdidae	*Turdus fumigatus*	Sabiá-da-mata	LC	Pet
	Turdus leucomelas	Sabiá-barranco	LC	Pet
	Turdus nudigenis	Caraxué	LC	Pet
MAMMALS				
Aotidae	*Aotus nancymai*	Macaco-da-noite	LC	Laboratory
	Aotus vociferans	Macaco-da-noite	LC	Laboratory
	Aotus nigriceps	Macaco-da-noite	LC	Laboratory
Atelidae	*Alouatta seniculus juara*	Bugio-vermelho-do-Juruá	LC	Food
	Alouatta seniculus	Guariba-vermelho	NL	Food
	Alouatta belzebul	Guariba-de-mãos-vermelhas	VU	Medicinal
	Alouatta nigerrima	Bugio-preto	LC	Medicinal
	Lagothrix lagothricha	Macaco-barrigudo	VU	Food
Balaenopteridae	*Balaenoptera acutorostrata*	Baleia-de-minke	LC	Medicinal
Bradypodidae	*Bradypus variegatus*	Bicho-preguiça	LC	Medicinal, Magical-religious
	Bradypus tridactylus	Preguiça-bentinho	LC	Medicinal
Canidae	*Cerdocyon thous*	Cachorro-do-mato	LC	Magical-religious
Caviidae	*Hydrochoerus hydrochaeris*	Capivara	LC	Food, Medicinal
Cebidae	*Sapajus apella*	Macaco-prego	LC	Medicinal
Cervidae	*Mazama americana*	Veado-mateiro	DD	Food, Medicinal, Magical-religious
	Mazama sp.	Veado	NL	Food, Medicinal
	Mazama gouazoubira	Veado-catingueiro	LC	Food
	Ozotoceros bezoarticus	Veado-campeiro	VU	Medicinal, Magical-religious
Cuniculidae	*Cuniculus paca*	Paca	LC	Food, Medicinal, Magical-religious
Dasypodidae	*Dasypus novemcinctus*	Tatu-galinha	LC	Food, Medicinal, Magical-religious
	Priodontes maximus	Tatu-canastra	VU	Food
	Dasypus sp.	Tatu	NL	Food
	Euphractus sexcinctus	Tatu-peba	LC	Medicinal, Magical-religious
	Tolypeutes sp.	Tatu-bola	NL	Medicinal
Dasyproctidae	*Dasyprocta fuliginosa*	Cutia	LC	Food
	Dasyprocta sp.	Cutia	NL	Magical-religious
Delphinidae	*Sotalia fluviatilis*	Tucuxi	NT	Medicinal, Magical-religious, Ornamental
	Sotalia guianensis	Boto-cinza	VU	Medicinal, Magical-religious, Ornamental
Didelphidae	*Didelphis marsupialis*	Gambá	LC	Medicinal, Magical-religious, Food
Erethizontidae	*Coendou prehensilis*	Porco-espinho	LC	Medicinal, Magical-religious
Felidae	*Panthera onca*	Onça	VU	Magical-religious
	Puma concolor	Onça-parda, Suçuarana	VU	Magical-religious
Iniidae	*Inia geoffrensis*	Boto-rosa	EN	Medicinal, Magical-religious
Mustelidae	*Eira barbara*	Irara	LC	Food
Myrmecophagidae	*Myrmecophaga tridactyla*	Tamanduá-bandeira	VU	Magical-religious
Physeteridae	*Physeter catodon*	Cachalote	VU	Medicinal
Procyonidae	*Procyon cancrivorus*	Guaxinim	LC	Medicinal, Magical-religious
	Nasua nasua	Quati	LC	Medicinal, Magical-religious
Tapiridae	*Tapirus terrestris*	Anta	VU	Food, Medicinal
Tayassuidae	*Tayassu pecari*	Queixada, Porco-do-mato	VU	Food, Medicinal
	Pecari tajacu	Catitu, Cateto	LC	Food
Trichechidae	*Trichechus inunguis*	Peixe-boi-da-Amazônia	VU	Food, Medicinal
	Trichechus manatus	Peixe-boi-marinho	EN	Medicinal

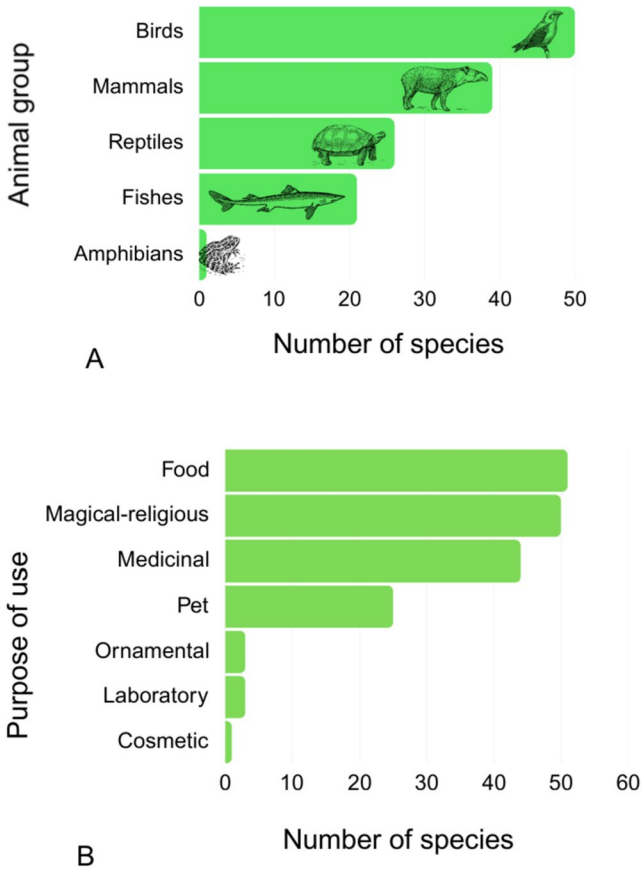


Fig. 4 Characterization of wildlife illegally traded in the Brazilian Amazon, according to the reviewed literature. (A) Number of species recorded for each animal group ($n=137$); (B) purposes of use for the traded species

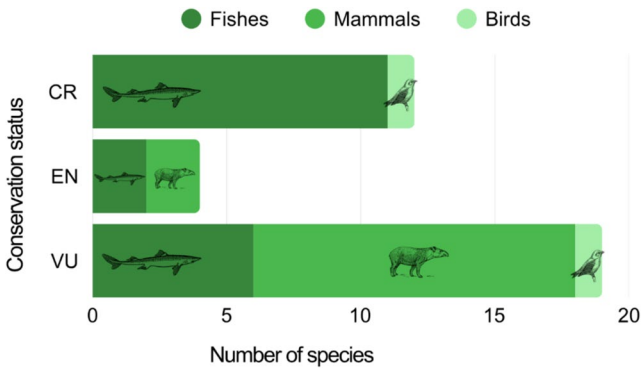


Fig. 5 Conservation status of species illegally traded in the Brazilian Amazon, by taxonomic group, according to the Red Book of Threatened Brazilian Fauna (ICMBio/MMA, 2018). CR: critically endangered; EN: endangered; VU: vulnerable

human consumption (e.g., *Podocnemis unifilis*; Pantoja-Lima et al., 2014), primates for laboratory use (e.g., *Aotus nigriceps*; Maldonado et al., 2009), and pet birds (e.g., *Sporophila angolensis*; Farias et al., 2019).

Illegal Wildlife Trade Routes in the Brazilian Amazon

Trade routes reported in the articles occurred at both national and international levels, with some articles documenting routes at both scales ($n=2$). National routes were mentioned in 12 articles (35.3%) and included movements between municipalities ($n=14$) and between states ($n=3$: Amazonas, Amapá, Pará) (Fig. 6). One such case is the transport of wild meat such as capybara and paca from the state of Amazonas to be sold at markets in Pará (Baía Júnior et al., 2010).

International routes were reported in five articles (14.7%) involving the trade of animals from Brazil to neighboring countries: Colombia, Guyana, French Guiana, Peru, and Suriname. Some examples included the transport of caiman meat from the Tefé region, in the central mesoregion of the state of Amazonas, to Colombia (da Silveira & Thorbjarnarson, 1999), and pet birds from inland Pará to be resold in Suriname, French Guiana, and Guyana (Silva et al. 2022a).

Drivers of IWT in the Brazilian Amazon

Among the articles analyzed, 18 (52.9%) reported drivers of IWT, which we classified as cultural, biological, socioeconomic, or environmental (Table 2), with several articles citing two or more factors. Cultural factors were the most frequently reported ($n=12$, 66.6%), with terms such as "preservation of heritage," "family cultural habit," and "cultural heritage" invoked to justify the use of animals for food, medicine, and pets. For example, the family tradition of raising and trading pet birds (Silva et al. 2022a). Biological factors were mentioned in 11 articles (61.1%) and include animal characteristics such as size ($n=5$, 27.7%), flavor of meat ($n=5$, 27.7%), and appearance and behavior of the animals ($n=1$, 5.5%). For instance, the size of animals used for human consumption influences choice, with a preference for larger animals (da Silveira & Thorbjarnarson, 1999).

Among socioeconomic factors, the price of animal products was identified as a driver of IWT for medicinal use and human consumption ($n=6$, 33.3%). One example is the lower price of remedies made from wild animals compared to pharmaceutical medicines sold in pharmacies (Alves & Santana, 2008; Pinto & Maduro, 2003). Among environmental factors, seasonality ($n=5$, 27.7%) and high local biodiversity ($n=1$, 5.5%) were also reported. For example, the trade of pet birds in local markets in Abaetetuba (PA) is more intense during the summer, as the availability of birds increases in this season, favoring their capture and trade (Farias et al., 2019).

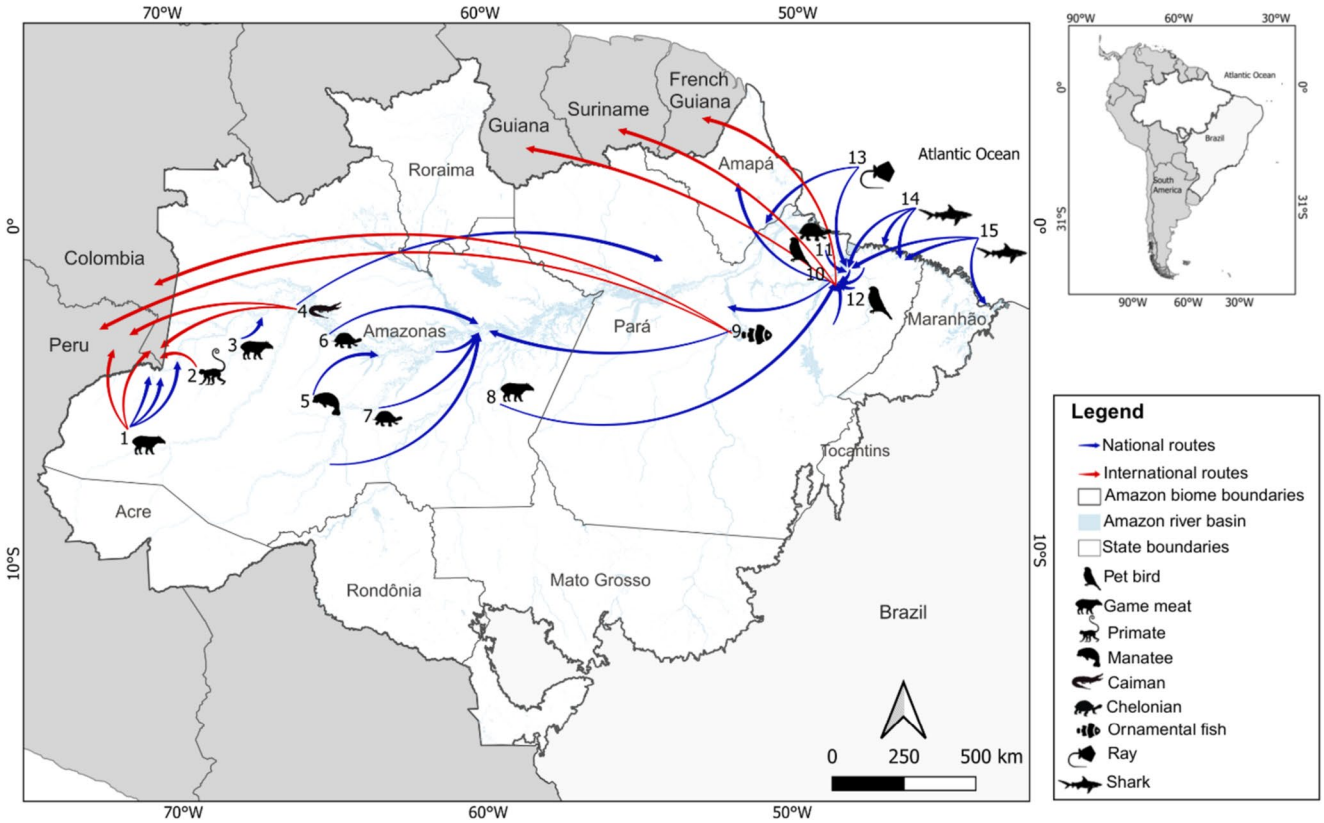


Fig. 6 IWT routes in the Brazilian Amazon recorded in the reviewed articles ($n=15$). Arrows represent the movement of animals from the extraction site to the trade location. Blue arrows indicate national trade routes, and red arrows indicate international trade routes. The figures represent the animal groups reported in the articles, and the numbers correspond to the reviewed articles evaluated: 1: van Vliet et al. 2015a, b; 2: Maldonado et al., 2009; 3: el Birzi et al., 2020; 4: da Silveira & Thorbjarnarson, 1999; 5: Franzini et al., 2013; 6: Johns, 1987; 7: Pantoja-Lima et al., 2014; 8: Baía Júnior et al., 2010; 9: Sousa et al., 2021; 10: Silva et al. 2022a; 11: Cristo et al., 2017; 12: Farias et al., 2019; 13: Filho et al., 2020; 14: Souza-Araujo et al., 2021; 15: Feitosa et al., 2018

Discussion

In this study, we conducted the first systematic review of the literature on IWT in the Brazilian Amazon, providing a broad and cohesive overview of this activity at a biome-wide scale. The data reveal that IWT is a widespread activity, often carried out in public places and involving a large diversity of species. The trade routes indicate the transport of animals over long distances within the national territory and highlight the existence of international trade. Moreover, the drivers of IWT observed reflect patterns similar to those documented in other regions of Brazil and in international contexts, suggesting a convergence of motivations and socio-environmental vulnerabilities associated with IWT.

The concentration of research conducted in the states of Pará and Amazonas is consistent with the greater infrastructure of universities and research centers (McManus et al., 2021) and the high occurrence of IWT in these states (Baía Júnior et al., 2010; van Vliet et al., 2014; Mendes, 2020). Belém (state of Pará) hosts one of the largest open-air markets in Brazil, where products from Amazonian fauna and flora are widely traded (Silva, 2011). In addition, the city is recognized as a center for acquiring medicinal products and animals, supplying markets in other states (Alves et al., 2010). The prevalence of studies in municipalities located along the main rivers may reflect the relative ease of access and the concentration of the largest urban and research centers in these areas (dos Santos et al., 2015). On the other hand, the more remote regions of the biome were underrepresented, likely due to access difficulties, as factors such as logistics, proximity to research centers, and transportation availability negatively influence the execution of ecological studies in the Amazon (Carvalho et al., 2023). However, this spatial distribution might represent a structural bias in the scientific literature on IWT in the Brazilian Amazon, since the available evidence reflects the trade where researchers are present or able to access, not necessarily where it is most intense or diverse. Thus, as the actual dynamics of trade remain poorly understood to both science and enforcement agencies, conservation policies risk being inadequate precisely where IWT is hardest to monitor. Additionally, restricting the search to peer-reviewed literature may have

Table 2 Drivers of IWT in the Brazilian Amazon according to the reviewed literature, with descriptions of each factor type and bibliographic references organized by purpose of wildlife use. The inclusion criteria for each driver are provided in Supplementary Material S5

Factor	Type	Description	Purpose of use	References
Cultural	Cultural habits	Cultural heritage of consuming or using wild animals and/or its parts/products.	Food	[7; 8; 9; 11; 20; 22; 32; 33]
			Medicinal	[2; 23]
			Pet	[12; 27]
Biological	Flavor	The distinctive flavor of wild meat drives demand and consumption of wild animals.	Food	[7; 8; 20; 28; 33]
	Size	Preference for larger animals, as they yield more meat.		[33; 21; 11; 17; 9]
	Appearance and behavior	The characteristics of birds (e.g., coloration, size, vocalization) influence the selection of species traded.	Pet	[27]
Socioeconomic	Price	Products derived from wild animals (e.g., meat, medicines) are priced lower than conventional products.	Food	[7; 20; 22]
			Medicinal	[3; 2; 23]
Environmental	Seasonality	The amount of wild meat traded increases during the rainy season due to a decrease in available fish.	Food	[33; 32]
		The trade of turtles and turtle eggs rises in the summer due to nesting.		[22]
		The trade of caimans increases in the summer, as they are easier to capture when rivers dry up.		[9]
		The trade of pet birds increases in the summer because of their greater availability and activity.	Pet	[12]
	Local biodiversity	Due to the high diversity of fauna, there is a wide variety of remedies made from wild animals.	Medicinal	[1]

limited the geographic and temporal coverage of the findings, particularly for regions and periods less represented in academic literature, which may be more frequently documented in gray literature and other non-academic sources.

The high incidence of illegal trade in public-access locations may be an indication of the strong demand for wild animals and their products (Charity & Ferreira, 2020), especially for wild meat (van Vliet et al., 2014; el Birzi et al., 2020). The same occurs in the Peruvian Amazon, where urban population growth drives demand for wild meat and the maintenance of fairs and public markets, including the Belén Market in the city of Iquitos, considered the largest wildlife market in Peru (Mayor et al., 2021). Furthermore, as suggested by Brookes et al. (2022) and Rivera et al. (2024), this pattern may be linked to the absence of regulation or inefficient enforcement of laws. Even in countries where trade is legally prohibited (e.g., Laos, India), the lack of monitoring and punishment facilitates the continuation of illegal trade without major consequences for traders (Brookes et al., 2022). In Brazil, although protective legislation for wildlife exists (Brazil, 1967), the enforcement of such laws and monitoring remain quite deficient, especially in the Amazon, due to the vast territorial scale and logistical factors (Charity & Ferreira, 2020). Meanwhile, forms of trade that occur outside public spaces have the advantage of evading potential monitoring, since they do not have a fixed point of sale and can be easily relocated (Farias et al., 2019).

The high representation of illegally traded fish species likely reflects environmental factors, as well as the region's dietary and cultural habits (da Costa et al., 2013). The Amazon Basin harbors the greatest diversity of freshwater fish on the planet (Dagosta & Pina, 2019), facilitating the availability, access, and traditional consumption of this resource, with the Amazon region having the highest fish consumption rate in the country (Lopes et al., 2016). Furthermore, fish consumption has cultural roots that connect it to local identity and tradition (da Costa et al., 2013). Mammals and birds were also widely traded for human consumption, reinforcing that the trade and consumption of wild meat are common and important practices in the Amazon (Morsello et al., 2015; van Vliet et al., 2018; Nunes et al., 2019; Torres et al., 2022; Aragão-Silva et al., 2023). This same consumption pattern is observed in other tropical regions, such as Africa and Asia, where wild animals, especially mammals, are traded as food (Greengrass et al., 2016; Gray et al., 2018; Brookes et al., 2022). In Brazil, a recent systematic review on hunting and wild meat consumption (Aragão-Silva et al., 2023) pointed out that the consumption of wild birds is concentrated in the North and Northeast regions of the country, with the greatest diversity of bird families being traded in the North. As observed in our review, these are mainly medium- and large-bodied species such as curassows (*Crax* spp.), black-bellied whistling ducks (*Dendrocygna autumnalis*), and guans (*Penelope spp.*) (van Vliet et al., 2015a; Silva et al. 2022b). Do Nascimento et al. (2015) found that in the state of Amazonas, food consumption was the main

motivation for the illegal trade of wild birds, and that this usage pattern differed from other regions of the country, where the main use is as pets (Destro et al., 2012).

Although some studies cited the pet trade, highlighting birds as the main animal group, this purpose of use was not as prominent among the reviewed articles. This result is similar to that observed by D'Cruze et al. (2024) in public markets in the Peruvian Amazon, suggesting that in the Amazon, the demand for consumptive uses of wildlife is higher than the demand for recreational or non-consumptive uses (Prasad et al., 2022). On the other hand, the diversity of species used for medicinal and magical-religious purposes observed in our review may be related to local beliefs regarding the efficacy of wild animals in treating human diseases (Alves & Rosa, 2007), as well as the diversity of religious and mystical practices in the Amazon region (Alves & Filho, 2007; Lima & Nobre, 2023). Amphibians, for instance, were recorded in our review only for medicinal purposes, despite being predominantly traded as pets on a global scale (Mohanty & Measey, 2019; Hughes et al., 2021).

The animal trade routes, both national and international, indicate the existence of an active trafficking network, considerable demand for wildlife products, and weak law enforcement (Charity & Ferreira, 2020). While individual studies have documented trade routes on a local scale, this review brings together evidence that these routes form part of a broader and structured trafficking network, a pattern that only becomes visible when evidence is systematically integrated across studies. The results indicate that wildlife is sourced from conserved and remote regions to supply urban markets with high demand and possible scarcity of these animals (Johns, 1987; Pantoja-Lima et al., 2014; Cristo et al., 2017). The international trade in animals highlights the existence of vulnerable border regions within the Amazon biome (Malamud, 2018), especially the Brazil–Colombia–Peru tri-border area, recognized as an important corridor for wildlife trafficking and its derivatives (Maldonado et al., 2009; van Vliet et al., 2018). The case of *Hypancistrus zebra* (Loricariidae), a fish endemic to the Xingu River, 2,000 km from the tri-border area, illustrates the high international demand for the species and a trafficking network that crosses the border (Beltrão et al., 2021; Sousa et al., 2021). Considering that the transport of the species occurs via waterways and air routes, strengthening surveillance at ports, airports, and borders is recommended to mitigate the trafficking of this species. On the other hand, the international trade in passerines to Guyana, Suriname, and French Guiana suggests that this region serves as a strategic route for bird trafficking from Brazil (Charity & Ferreira, 2020). A delicate aspect of illegal trade in this region is the specific legislation each country adopts regarding wildlife trade: while Brazil has had its wildlife protected by law since 1967 (Law No. 5,197/1967), Guyana and Suriname have a more liberal approach to the commercial use of wildlife (Ortiz-Von, 2018). These legislative gaps clearly pose pressure and threats to species that may be captured in Brazilian territory and resold legally in neighboring countries (Charity & Ferreira, 2020).

Regarding the drivers of IWT, this review provides the first systematic identification and categorization of the factors motivating illegal wildlife trade in the Brazilian Amazon. Cultural factors have been widely documented as key motivations for the trade and consumption of wild meat in other tropical regions, such as Central Africa (van Vliet & Mbazza, 2011). Despite its cultural and dietary importance, this practice, when carried out unsustainably, contributes to the decline of biodiversity and animal populations (Gardner et al., 2019). Threatened species such as the lowland tapir (*Tapirus terrestris*), giant armadillo (*Priodontes maximus*), and white-lipped peccary (*Tayassu pecari*) were recorded in our study as commercially traded, suggesting they may face additional pressure from illegal hunting, though the direct impact of trade on their populations remains to be assessed. More effective enforcement, together with awareness-raising initiatives about the impacts of illegal trade on species, could help mitigate this threat. In many cases, cultural motivations are difficult to disentangle from socioeconomic conditions, as cultural traditions and economic vulnerability frequently reinforce each other in sustaining the demand for wildlife products (Morsello et al., 2015; Nielsen et al., 2018).

In terms of socioeconomic factors, the demand for wild meat and natural remedies, due to their lower cost compared to conventional products, is frequently observed (Hughes et al., 2022), particularly in low-income contexts (Duffy et al., 2016). Furthermore, illegal wildlife fairs or trade represent a source of livelihood for many people (e.g., da Silveira & Thorbjarnarson, 1999; Maldonado et al., 2009; van Vliet et al., 2015a; Cristo et al., 2017; Farias et al., 2019; el Birzi et al., 2020; Silva et al. 2022a), which contributes to the persistence of these markets. In such contexts, culturally rooted practices such as wild meat consumption are not only maintained but also reinforced by economic necessity (Nielsen et al., 2018), strengthening the continuity of illegal trade. Addressing these drivers therefore requires improvements in living conditions, including better work and educational opportunities for vulnerable populations. In this context, alternatives such as community-based management of species, for example, the pirarucu (*Arapaima gigas*) (Campos-Silva & Peres, 2016), are strategies that can serve as models for generating income in a sustainable way.

Regarding biological factors, studies show that the preference for larger animals is related to the greater amount

of meat available and, in some cases, to the greater likelihood of sale (da Silveira & Thorbjarnarson, 1999; Morcatty & Valsecchi, 2015; Hughes et al., 2022), given that larger individuals command higher market prices, reinforcing the economic dimension of this biological preference. This preference for larger animals was also observed by Mayor et al. (2021) when assessing wildlife trade in the Peruvian Amazon. Thus, the presence of medium- and large-sized species such as the paca (*Cuniculus paca*), lowland tapir (*Tapirus terrestris*), and peccary (*Tayassu pecari*) also indicates the availability and exploitation of this resource in the Amazon region. This scenario differs, for example, from the Brazilian semiarid region, where, due to the population decline of historically hunted mammal species, hunting pressures have shifted toward birds and small mammals (Mendonça et al., 2016; da Silva et al., 2020). Although not directly documented in the studies reviewed here, this pattern may serve as a cautionary analogy for the Amazon region, where hunting pressure and wildlife trade, together with other threats such as habitat fragmentation (Peres, 2001; Vliet et al., 2014), could similarly compromise the availability of medium- and large-sized species for populations that depend on this resource. Finally, environmental factors also represent an important dimension of IWT in the Amazon. Seasonal fluctuations in water levels and rainfall, documented in the region (Sorribas et al., 2016; Bottino et al., 2024), directly influence the intensity and accessibility of wildlife trade (Souza et al., 2024), representing a structurally embedded environmental dimension of IWT in the Amazon. However, the potential effects of these seasonal fluctuations on wildlife consumption and trade in the region are still poorly understood, and further research in this area is recommended.

Conclusion

By providing the first systematic review of IWT in the Brazilian Amazon, this study establishes a scientific foundation for understanding the scope and complexity of this phenomenon across the entire biome. The geographic and thematic gaps identified underscore the urgency of expanding research into remote and border areas, where IWT dynamics remain poorly understood.

It is important to note that the data analyzed in this study are based exclusively on published literature, which may underestimate the actual magnitude of IWT in the region. In addition, there is a growing trend of illegal trade migrating to virtual environments, such as social media, messaging groups, and online sales platforms (Borges et al., 2021; Wyatt et al., 2022). This strategy has been widely adopted by traffickers to evade enforcement and expand the reach of transactions, making tracking even more challenging (Demeau et al., 2019). Such dynamics reinforce the need to develop complementary monitoring methodologies capable of capturing the nuances of digital trade and increasing the visibility of practices that remain poorly documented.

The findings of this review provide a basis for the development of integrated policies combining environmental education, enforcement, and economic alternatives, tailored to the sociocultural realities of the Brazilian Amazon. Adapting demand-reduction strategies to the local Brazilian context, targeting both consumers and the general public, may represent an effective approach to mitigate IWT in the region (Burgess, 2016; Olmedo et al., 2021). Furthermore, the development of sustainable economic alternatives for local populations can contribute significantly to the conservation of Amazonian biodiversity. At the same time, it is crucial to strengthen enforcement and control of illegal trade at ports, airports, rivers, urban markets, and fairs, as well as to improve the allocation of financial and human resources to enforcement agencies. Combating international trade routes requires cooperation among Amazonian countries, as a lack of coordination between enforcement agencies and police forces undermines individual control efforts (Charity & Ferreira, 2020). Moreover, understanding the challenges and perspectives of the agencies responsible for enforcement in the Amazon is essential, making research in this field indispensable.

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Author contributions MCL conceived the idea, performed the literature search, and, together with MFDF, designed the methodology and carried out the analyses. MFDF, RRNA, and NS provided substantial input through critical review, editing, and intellectual contributions to the manuscript. All authors read and approved the final version.

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethics Statements Not applicable.

Clinical Trial Number Not applicable.

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